Ex 4.3.1

Use the Fundamental Theorem for Path Integrals to show that

$$\int_{\partial\Delta(-1-i,1-i,i)} \frac{1}{z}\,dz = 2\pi i.$$

(Hint: the principal branch of $\log(z)$ is an antiderivative for $1/z$ in $D^* = \mathbb{C} \setminus (-\infty, 0]$. But we can also define a branch of $\log(z)$ in $\mathbb{C} \setminus [0, \infty)$ by $g(z) = \ln|z| + i\arg_0(z)$ where $\arg_0(z) \in (0, 2\pi)$.)

Pf. Let $A = -1 - i$, $B = 1 - i$, $C = i$ be the vertices. The boundary $\partial\Delta$ is over $[A, B] + [B, C] + [C, A]$.

The principal branch $\mathrm{Log}(z) = \ln|z| + i\,\mathrm{Arg}(z)$ with $\mathrm{Arg}(z) \in (-\pi, \pi]$ is an antiderivative for $1/z$ on $D^* = \mathbb{C}\setminus(-\infty, 0]$. The segment $[A, B]$ has $\mathrm{Im}(z) = -1$ throughout, so it avoids $(-\infty, 0]$. The segment $[B, C]$ from $1 - i$ to i crosses the real axis only at $z = \frac{1}{2} > 0$, so it also avoids $(-\infty, 0]$. Hence both edges lie in D^*, and by the Fundamental Theorem for Path Integrals, we can write

$$\int_{[A,B]+[B,C]} \frac{dz}{z} = \mathrm{Log}(C) - \mathrm{Log}(A).$$

Then we can evaluate,

$$\mathrm{Log}(i) = \ln 1 + i\frac{\pi}{2} = \frac{\pi i}{2}$$
$$\mathrm{Log}(-1 - i) = \ln\sqrt{2} + i\left(-\frac{3\pi}{4}\right) = \tfrac{1}{2}\ln 2 - \tfrac{3\pi i}{4}$$

Therefore,

$$\mathrm{Log}(C) - \mathrm{Log}(A) = \frac{\pi i}{2} - \frac{1}{2}\ln 2 + \frac{3\pi i}{4} = -\frac{1}{2}\ln 2 + \frac{5\pi i}{4}$$

The branch $g(z) = \ln|z| + i\arg_0(z)$ with $\arg_0 \in (0, 2\pi)$ is an antiderivative for $1/z$ on $\mathbb{C} \setminus [0, \infty)$. The segment $[C, A]$ from i to $-1 - i$ crosses the real axis only at $z = -\frac{1}{2} < 0$, so it avoids $[0, \infty)$. By the Fundamental Theorem of Path Integrals again, we write

$$\int_{[C,A]} \frac{dz}{z} = g(A) - g(C).$$

Since they are in the third quadrant, we have $\arg_0(i) = \pi/2$ and $\arg_0(-1 - i) = 5\pi/4$. Then,

$$g(i) = \frac{\pi i}{2}$$
$$g(-1 - i) = \tfrac{1}{2}\ln 2 + \tfrac{5\pi i}{4}.$$

Therefore,

$$g(A) - g(C) = \frac{1}{2}\ln 2 + \frac{5\pi i}{4} - \frac{\pi i}{2} = \frac{1}{2}\ln 2 + \frac{3\pi i}{4}.$$

Summing these we get,

$$\int_{\partial\Delta} \frac{dz}{z} = \left(-\frac{1}{2}\ln 2 + \frac{5\pi i}{4}\right) + \left(\frac{1}{2}\ln 2 + \frac{3\pi i}{4}\right) = \frac{8\pi i}{4}$$

which simplifies to $2\pi i$. □

Ex 4.2.1

Suppose that f is holomorphic in a domain D containing $\overline{D_r(z_0)}$ for some $r > 0$. Show that

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta.$$

Thus, $f(z_0)$ is the average of the values of f on the circle $|z - z_0| = r$.

Pf. We have that D is open and contains the compact set, $\overline{D_r(z_0)}$. Then there exists an $R > r$ such that $D_R(z_0) \subset D$. The disk $D_R(z_0)$ is a convex domain with $f \in H(D_R(z_0))$.

Let $\gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$. From Lemma 4.4, we have $\mathrm{Ind}_\gamma(z_0) = 1$ (since $|z - z_0| < r$). We also have $\gamma^* \subset D_R(z_0)$. From Theorem 4.10 (Cauchy Integral) on the convex domain $D_R(z_0)$, we get

$$\begin{aligned} f(z_0) &= \mathrm{Ind}_\gamma(z_0) f(z_0) \\ &= \frac{1}{2\pi i} \int_\gamma \frac{f(w)}{w - z_0}\, dw. \end{aligned}$$

We can parametrize, $w = z_0 + re^{i\theta}$, $dw = ire^{i\theta}\, d\theta$, $w - z_0 = re^{i\theta}$. Then substitute to integrate,

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{re^{i\theta}} ire^{i\theta}\, d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta \end{aligned}$$

which shows $f(z_0)$ is the average of f over the circle $|z - z_0| = r$. □

Ex 4.3.2

Let D be a domain and $p \in D$. Suppose that f is continuous on D and that $f \in H(D \setminus \{p\})$. Show that $f \in H(D)$.

Pf. Since D is open and $p \in D$, $\exists r > 0$ s.t. $D_r(p) \subset D$.

The disk $D_r(p)$ is a convex domain. On $D_r(p)$ we have: f is continuous and $f \in H(D_r(p) \setminus \{p\})$. These are exactly the conditions in Theorem 4.8 (Cauchy for Convex Domains) with exceptional point p.

By Theorem 4.8, f has an antiderivative F in $D_r(p)$. Then, $F \in H(D_r(p))$ and $F' = f$ on $D_r(p)$.
By Corollary 4.13, $f = F' \in H(D_r(p))$. In particular, f is holomorphic at p. Since f was already holomorphic on $D \setminus \{p\}$, we have shown that $f \in H(D)$. □

Ex 4.3.4

Suppose that $f(z) = u(z) + iv(z)$ with u and v real-valued. If D is open and $f \in H(D)$, show that u and v are smooth on D in the sense that they have continuous partial derivatives of all orders. In particular, conclude that u and v are harmonic on D. (Compare with Ex 2.2.8.)

Pf. Since $f \in H(D)$, Theorem 4.12 (Holomorphic Functions are Analytic) gives f is analytic on D. By Corollary 4.13, f has holomorphic (hence continuous) derivatives of all orders: $f^{(k)} \in H(D)$ for every $k \geqslant 0$.

By equation (2.9) of Theorem 2.14, wherever $f'(z)$ exists we have

$$f'(z) = f_x(z) = u_x(z) + iv_x(z).$$

Since $f' \in H(D)$, f' is continuous, so $u_x = \mathrm{Re}(f')$ and $v_x = \mathrm{Im}(f')$ are continuous on D.

The Cauchy–Riemann equations $u_y = -v_x$ and $v_y = u_x$ then give that u_y and v_y are also continuous. Thus all first-order partial derivatives of u and v are continuous on D. Thus $u, v \in C^1(D)$.
We can prove by induction that for $f^{(k-1)}$ for each k, we obtain $u, v \in C^k(D)$ $\forall k$. The same reasoning applies to f' (which is itself holomorphic), then f'' is continuous, so $(u_x)_x$ and $(v_x)_x$ are continuous. By Cauchy–Riemann again for f', all second-order partial derivatives of u and v are continuous, giving $u, v \in C^2(D)$, and so on.
Hence u and v are smooth (C^∞) on D.

Since $u, v \in C^2(D)$, we differentiate the Cauchy–Riemann equations:

$$\begin{aligned} u_{xx} &= (v_y)_x = v_{yx} \\ u_{yy} &= (-v_x)_y = -v_{xy}. \end{aligned}$$

Since v is C^2, by Clairut we have $v_{yx} = v_{xy}$. Therefore,

$$u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$$

and,

$$\begin{aligned} v_{xx} &= (-u_y)_x = -u_{yx} \\ v_{yy} &= (u_x)_y = u_{xy}, \end{aligned}$$

thus $v_{xx} + v_{yy} = -u_{yx} + u_{xy} = 0$. Hence shown that u and v are harmonic on D. □

Ex 4.3.8

Let

$$f(z) = \frac{1}{e^z + 1} = \sum_{n=0}^{\infty} a_n z^n$$

be the MacLaurin series for f.

(a) Use Ex 4.3.7 to show that the a_n satisfy $a_0 = \frac{1}{2}$ and

$$a_n = -\frac{1}{2} \sum_{k=0}^{n-1} \frac{a_k}{(n-k)!} \quad \text{if } n \geqslant 1.$$

(b) Use part (a) to calculate the first few terms of the series.

(c) What is the radius of convergence of the series?

Pf.

(a) Since $f(z)(e^z + 1) = 1$, we expand both factors as power series and use the Cauchy product (from Ex 4.3.7).

The MacLaurin series for $e^z + 1$ is

$$\sum_{n=0}^{\infty} b_n z^n$$

where $b_0 = 2$ and $b_n = 1/n!$ for $n \geqslant 1$.

From Ex 4.3.7 we have,

$$f(z)(e^z + 1) = \sum c_n z^n$$

where $c_n = \sum_{k=0}^{n} a_k \, b_{n-k}$. Since the product equals the constant 1, we have $c_0 = 1$ and $c_n = 0$ for $n \geqslant 1$.

From $c_0 = a_0 b_0 = 2a_0 = 1$, we get $a_0 = \frac{1}{2}$.

For $n \geqslant 1$, we have,

$$\begin{aligned} 0 = c_n &= a_n b_0 + \sum_{k=0}^{n-1} a_k \, b_{n-k} \\ &= 2a_n + \sum_{k=0}^{n-1} \frac{a_k}{(n-k)!}. \end{aligned}$$

Then we have, $a_n = -\frac{1}{2} \sum_{k=0}^{n-1} \frac{a_k}{(n-k)!}$ for $n \geqslant 1$. □

(b) Using the pattern from part (a), we have,

$$\begin{aligned} a_0 &= \tfrac{1}{2} \\ a_1 &= -\tfrac{1}{2} \cdot \frac{a_0}{1!} = -\tfrac{1}{4} \\ a_2 &= -\tfrac{1}{2}\Big(\frac{a_0}{2!} + \frac{a_1}{1!}\Big) = -\tfrac{1}{2}\big(\tfrac{1}{4} - \tfrac{1}{4}\big) = 0 \\ a_3 &= -\tfrac{1}{2}\Big(\frac{a_0}{3!} + \frac{a_1}{2!} + \frac{a_2}{1!}\Big) = -\tfrac{1}{2}\big(\tfrac{1}{12} - \tfrac{1}{8} + 0\big) = -\tfrac{1}{2}\big(-\tfrac{1}{24}\big) = \tfrac{1}{48} \\ a_4 &= -\tfrac{1}{2}\Big(\frac{a_0}{4!} + \frac{a_1}{3!} + \frac{a_2}{2!} + \frac{a_3}{1!}\Big) = -\tfrac{1}{2}\big(\tfrac{1}{48} - \tfrac{1}{24} + 0 + \tfrac{1}{48}\big) = 0 \end{aligned}$$

Thus $f(z) = \frac{1}{2} - \frac{1}{4} z + \frac{1}{48} z^3 + \cdots$ (where even coeffs zero out).

(c) The function $f(z) = 1/(e^z + 1)$ has singularity when $e^z = -1$, i.e. $z = (2k+1)\pi i \; \forall k \in \mathbb{Z}$.

The closest singularities to origin are $z = \pm \pi i$, both having distance of π. By Corollary 4.14 (Taylor's Theorem), the radius of convergence is at least π. Since f is not holomorphic at $z = \pi i$, the power series cannot converge after distance π. Therefore the radius of convergence is exactly π.

EP-2

Complete the proof of Cauchy's Theorem for Convex Domains by showing that if f is continuous at z_0, then

$$\lim_{z \to z_0} \frac{1}{z - z_0} \int_{[z_0,\, z]} \big(f(w) - f(z_0)\big)\, dw = 0.$$

Pf. From equation (4.3) in Theorem 4.8 (Cauchy for Convex Domains), given $F(z) = \int_{[a,z]} f(w)\, dw$ and Cauchy's Theorem for a Triangle, we have

$$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0} \int_{[z_0,\, z]} \big(f(w) - f(z_0)\big)\, dw.$$

We can show that the RHS tends to 0 as z tends to z_0.

Let $\varepsilon > 0$. Since f is continuous at z_0, $\exists \delta > 0$ s.t.

$$|w - z_0| < \delta \implies |f(w) - f(z_0)| < \varepsilon$$

$\forall w$ on segment $[z_0, z]$, we have $|w - z_0| \leqslant |z - z_0|$. So when $|z - z_0| < \delta$, every $w \in [z_0, z]$ satisfies $|f(w) - f(z_0)| < \varepsilon$. By Proposition 3.26(a) to estimate the standard path-integral, we have,

$$\left| \int_{[z_0,\, z]} \big(f(w) - f(z_0)\big)\, dw \right| \leqslant \varepsilon \ell\big([z_0, z]\big) = \varepsilon\, |z - z_0|$$

$$\left| \frac{1}{z - z_0} \int_{[z_0,\, z]} \big(f(w) - f(z_0)\big)\, dw \right| \leqslant \varepsilon$$

Since we have an arbitrary $\varepsilon > 0$, limit tends to 0. □

Ex 5.1.2

Suppose that D is a domain and that $f \in H(D)$ is nonconstant. Suppose also that there is an analytic branch of $f(z)^{1/n}$ in D. (That is, there is a $g \in H(D)$ such that $f(z) = g(z)^n$.) Show that if f has a zero of order m in D, then m is divisible by n.

Pf. Let $a \in D$ be a zero of f of order m. Since $f(a) = g(a)^n = 0$, we have $g(a) = 0$. Also, since f is nonconstant, g is nonconstant (otherwise $f = g^n$ would be constant).

By Theorem 5.6, g has a zero of finite order at a. Let this order be some $k \geqslant 1$.
Then $h \in H(D)$ with $h(a) \neq 0$, we have by Lemma 5.4 that

$$g(z) = (z-a)^k h(z)$$

$\forall z \in D$. Let $f(z)$ be the be $g(z)$ raised to the nth power s.t.

$$f(z) = g(z)^n = (z-a)^{nk}\, h(z)^n$$

Since we have $h(a) \neq 0$, the function $h(z)^n$ is holomorphic with $h(a)^n \neq 0$. By Corollary 2.7, we know that the order of a zero is unique, and thus we must have $m = nk$. Hence m is divisible by n. □

Ex 5.1.3

Suppose that f is entire and that

$$f\left(\frac{1}{n}\right) = f\left(-\frac{1}{n}\right) \quad \text{for all } n \in \mathbb{N}.$$

Show that $f(z) = f(-z)$ for all $z \in \mathbb{C}$.

Pf. Let $h(z) = f(z) - f(-z)$. Since f is entire and the map z to $-z$ is also trivially entire, we can show from Proposition 2.2 (composition of entire functions) that the composition $f(-z)$ is entire. Therefore, we have that h is entire as the difference of two entire functions.

By hypothesis, for every $n \in \mathbb{N}$,

$$h\left(\frac{1}{n}\right) = f\left(\frac{1}{n}\right) - f\left(-\frac{1}{n}\right) = 0$$

So the zero set $Z(h) = \{z \in \mathbb{C} : h(z) = 0\}$ contains the set $S = \{1/n : n \in \mathbb{N}\} = \{1, \frac{1}{2}, \frac{1}{3}, \ldots\}$.

Since we have that $\dfrac{1}{n}$ tends to 0 as n tends to ∞, the set S has 0 as limit point. Hence $Z(h)$ has a limit point in the domain $\mathbb{C}$.

By Corollary 5.7, we have that $h = f(z) - f(-z)$ and zero function (both are holomorphic on $\mathbb{C}$) equal to each other in this set with limit point in domain $\mathbb{C}$, therefore they must equal each other everywhere, i.e. $h \equiv 0$.

Then, $0 \equiv f(z) - f(-z)$, which implies $f(z) = f(-z) \ \ \forall z \in \mathbb{C}$. □